

Mathematics - Upper Key Stage 2

Complete course material, four chapters.

Chapter 1: Place value and large numbers

Pupils read, write, order and compare numbers up to ten million. They determine the value of each digit and round any whole number to a required degree of accuracy. Negative numbers are introduced in context and pupils calculate intervals across zero. Practice exercises cover ordering, rounding and estimation.

Chapter 2: Fractions, decimals and percentages

Pupils compare and order fractions, including fractions greater than one. They add and subtract fractions with different denominators using the concept of equivalent fractions, multiply pairs of proper fractions, and divide proper fractions by whole numbers. Pupils recall and use equivalences between simple fractions, decimals and percentages, and solve problems involving the calculation of percentages.

Chapter 3: Geometry - properties of shapes

Pupils draw two dimensional shapes using given dimensions and angles, recognise and build simple three dimensional shapes including making nets, and compare and classify geometric shapes based on their properties and sizes. They find unknown angles in any triangles, quadrilaterals and regular polygons, and illustrate the parts of a circle including radius, diameter and circumference.

Chapter 4: Measurement and statistics

Pupils convert between miles and kilometres, calculate the area of parallelograms and triangles, and calculate the volume of cubes and cuboids. They interpret and construct pie charts and line graphs and use these to solve problems, and calculate and interpret the mean as an average of a set of data.